

1 Making Decisions under Uncertainty

2 Convergent Prior

We will use this notation for showing a random variable X with possible values from $\mathcal{X}$ and value x , a random vector $\mathbf{X}$ with range $\mathcal{X}$ and value $\mathbf{x}$, a set $\mathcal{A}$, and a measure $\mathbb{M}$.

Definition 2.1 (probabilistic model). We define a *probabilistic model* $\mathcal{M}$ as a pair $(\mathcal{S}, \mathcal{P})$, where $\mathcal{S}$ is a sample space, and $\mathcal{P}$ is a set of acceptable probability measures on $\mathcal{S}$.

Definition 2.2 (ϵ -convergence). Let $\mathbf{X}$ and $\mathbf{E}$ be two random vectors in a probabilistic model $\mathcal{M} \equiv (\mathcal{S}, \mathcal{P})$. We say $\mathbf{X}|\mathbf{E}$ ($\mathbf{X}$ given $\mathbf{E}$) ϵ -converges by a distribution P in $\mathcal{M}$, if for every distribution $Q \in \mathcal{P}$ we have:

$$\sum_{\mathbf{x} \in \mathcal{X}} \sum_{\mathbf{e} \in \mathcal{E}} Q(\mathbf{x}, \mathbf{e}) \ln \frac{Q(\mathbf{x} | \mathbf{e})}{P(\mathbf{x} | \mathbf{e})} < \epsilon . \quad (1)$$

We call P an ϵ -convergent distribution for $\mathbf{X}|\mathbf{E}$ in $\mathcal{M}$.

Definition 2.3 (convergence measure). Let $\mathbf{X}$ and $\mathbf{E}$ be random vectors in a probabilistic model $\mathcal{M} \equiv (\mathcal{S}, \mathcal{P})$. We define the *convergence measure* of $\mathbf{X}|\mathbf{E}$ by $P \in \mathcal{P}$ as follows:

$$\mathbb{C}_P [\mathbf{X}|\mathbf{E}] = \max_{Q \in \mathcal{P}} \sum_{\mathbf{x} \in \mathcal{X}} \sum_{\mathbf{e} \in \mathcal{E}} Q(\mathbf{x}, \mathbf{e}) \ln \frac{Q(\mathbf{x} | \mathbf{e})}{P(\mathbf{x} | \mathbf{e})} . \quad (2)$$

And, we define the convergence measure of $\mathbf{X}|\mathbf{E}$ in $\mathcal{M}$ as follows:

$$\mathbb{C}_{\mathcal{M}} [\mathbf{X}|\mathbf{E}] = \min_{P \in \mathcal{P}} \mathbb{C}_P [\mathbf{X}|\mathbf{E}] . \quad (3)$$

Proposition. If $\mathbb{C}_{\mathcal{M}} [\mathbf{X}|\mathbf{E}] \geq \epsilon$, then there is no ϵ -convergent distribution for $\mathbf{X}|\mathbf{E}$ in $\mathcal{M}$.

Definition 2.4 (parametric model). Let θ be a random vector. A parametric model is a probabilistic model $\mathcal{M}_{\theta} \equiv (\mathcal{S}, \mathcal{P})$, where for every $P \in \mathcal{P}$ there is a prior distribution p , such that for every $s \in \mathcal{S}$ we have:

$$P(s) = \int \pi(s|\theta)p(\theta)d\theta ,$$

while π is some known probability measure.

Theorem 1. Let $\mathcal{M}_\theta \equiv (\mathcal{S}, \mathcal{P})$ be a parametric model, and $\mathbf{X}, \mathbf{E}$ be two random vectors in $\mathcal{M}_\theta$. For every $P \in \mathcal{P}$ we have:

$$\mathbb{C}_P[\mathbf{X}|\mathbf{E}] < \max_{\theta \in \Theta} \sum_{\mathbf{x} \in \mathcal{X}} \sum_{\mathbf{e} \in \mathcal{E}} \pi(\mathbf{x}, \mathbf{e} | \theta) \ln \frac{\pi(\mathbf{x} | \mathbf{e}, \theta)}{P(\mathbf{x} | \mathbf{e})} . \quad (4)$$

Moreover, by bayes' theorem we have:

$$\frac{\pi(\mathbf{x} | \mathbf{e}, \theta)}{P(\mathbf{x} | \mathbf{e})} = \frac{P(\theta | \mathbf{x}, \mathbf{e})}{P(\theta | \mathbf{e})} . \quad (5)$$

Proof. □

Theorem 2 (Minimax). Let $X \subset \mathbb{R}^n$ and $Y \subset \mathbb{R}^m$ be compact convex sets. If $f : X \times Y \rightarrow \mathbb{R}$ is a continuous function that is concave-convex, i.e.

- $f(\cdot, y) : X \rightarrow \mathbb{R}$ is concave for fixed y , and
- $f(x, \cdot) : Y \rightarrow \mathbb{R}$ is convex for fixed x .

Then we have that:

$$\max_{x \in X} \min_{y \in Y} f(x, y) = \min_{y \in Y} \max_{x \in X} f(x, y) . \quad (6)$$

Theorem 3. Let $\mathbf{X}$ and $\mathbf{E}$ be two random vectors in a parametric model $\mathcal{M}_\theta \equiv (\mathcal{S}, \mathcal{P})$. For the convergence measure of $\mathbf{X}|\mathbf{E}$ in $\mathcal{M}_\theta$ we have:

$$\mathbb{C}_{\mathcal{M}_\theta}[\mathbf{X}|\mathbf{E}] < \max_{P \in \mathcal{P}} \int \sum_{\mathbf{x} \in \mathcal{X}} \sum_{\mathbf{e} \in \mathcal{E}} P(\mathbf{x}, \mathbf{e}, \theta) \ln \frac{\pi(\mathbf{x} | \mathbf{e}, \theta)}{P(\mathbf{x} | \mathbf{e})} d\theta , \quad (7)$$

where $P(\mathbf{x}, \mathbf{e}, \theta) = \pi(\mathbf{x}, \mathbf{e} | \theta) p(\theta)$.

Proof. □

Definition 2.5 (ϵ -convergent distribution). Let S be a random variable, and $\mathcal{E}$ be a set of random variables in a probabilistic model $\mathcal{M}$. We say P is an ϵ -convergent distribution for $S|\mathcal{E}$ in $\mathcal{M}$, if for all $E \in \mathcal{E}$, $S|E$ ϵ -converges by P in $\mathcal{M}$.

Proposition. Let S be a random variable, and $\mathcal{E}$ be a set of random variables in a probabilistic model $\mathcal{M}$. If there exists $E \in \mathcal{E}$ such that $\mathbb{C}_{\mathcal{M}}[S|E] \geq \epsilon$, then there is no ϵ -convergent distribution for $S|\mathcal{E}$ in $\mathcal{M}$,

Theorem 4. Consider two probabilistic models $\mathcal{M}$ and $\mathcal{M}^*$. Let $\mathcal{E}$ be a countable set of random variables in $\mathcal{M}$, and E^* be a single random variable in $\mathcal{M}^*$. Suppose that there exists a random variable I in $\mathcal{M}^*$, such that for **every** probability measure P in $\mathcal{M}$, there exists a probability measure P^* in $\mathcal{M}^*$, such that for every $E_i \in \mathcal{E}$, every $s \in \mathcal{S}$ and every $e_i \in \mathcal{E}_i$, there exist some values s^*, e^* such that we have:

$$P(S = s, E_i = e_i) = P^*(S^* = s^*, E^* = e^* \mid I = i) . \quad (8)$$

In addition, I and E^* are conditionally independent:

$$P^*(S^* \mid E^*) = P^*(S^* \mid E^*, I) . \quad (9)$$

Then $S|\mathcal{E}$ ϵ -converges by P in $\mathcal{M}$, if $S^*|E^*$ ϵ -converges by P^* in $\mathcal{M}^*$.

Proof. Assume, by contradiction, that $S|\mathcal{E}$ does not ϵ -converge by P in $\mathcal{M}$. Therefore, there must exist Q and $E_i \in \mathcal{E}$ in $\mathcal{M}$ such that:

$$\mathbb{E}_Q [\mathbb{D}(P_{S|E_i} \| Q_{S|E_i})] \geq \epsilon .$$

By using equations 8 and 9 we can rewrite the left hand side:

$$\begin{aligned} \mathbb{E}_Q [\mathbb{D}(P_{S|E_i} \| Q_{S|E_i})] &= \sum_{s \in \mathcal{S}} \sum_{e_i \in \mathcal{E}_i} \ln \frac{Q(s \mid e_i)}{P(s \mid e_i)} Q(s, e_i) \\ &= \sum_{s^* \in \mathcal{S}^*} \sum_{e^* \in \mathcal{E}^*} \ln \frac{Q^*(s^* \mid e^*, I = i)}{P^*(s^* \mid e^*, I = i)} Q^*(s^*, e^* \mid I = i) \\ &= \sum_{s^* \in \mathcal{S}^*} \sum_{e^* \in \mathcal{E}^*} \ln \frac{Q^*(s^* \mid e^*)}{P^*(s^* \mid e^*)} Q^*(s^*, e^* \mid I = i) . \end{aligned}$$

Hence, we have:

$$\sum_{s^* \in \mathcal{S}^*} \sum_{e^* \in \mathcal{E}^*} \ln \frac{Q^*(s^* \mid e^*)}{P^*(s^* \mid e^*)} Q^*(s^*, e^* \mid I = i) \geq \epsilon .$$

Since $Q^*(i)$ is a distribution and is non-negative for every i , we can multiply both sides of this inequality to $Q^*(i)$ and sum over all values of i :

$$\begin{aligned} \sum_i \sum_{s^* \in \mathcal{S}^*} \sum_{e^* \in \mathcal{E}^*} \ln \frac{Q^*(s^* \mid e^*)}{P^*(s^* \mid e^*)} Q^*(s^*, e^*, I = i) &\geq \sum_i \epsilon Q^*(i) \\ \sum_{s^* \in \mathcal{S}^*} \sum_{e^* \in \mathcal{E}^*} \ln \frac{Q^*(s^* \mid e^*)}{P^*(s^* \mid e^*)} \sum_i Q^*(s^*, e^*, I = i) &\geq \epsilon \sum_i Q^*(i) \\ \sum_{s^* \in \mathcal{S}^*} \sum_{e^* \in \mathcal{E}^*} \ln \frac{Q^*(s^* \mid e^*)}{P^*(s^* \mid e^*)} Q^*(s^*, e^*) &\geq \epsilon , \end{aligned}$$

which means $S^*|E^*$ does not ϵ -converge by P^* in $\mathcal{M}^*$. $\square$

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Definition 2.6 (conditional model). Let $\mathcal{M} \equiv (\mathcal{S}, \mathcal{P})$ be a probabilistic model. $\mathcal{M}$ conditioned on an event $\mathcal{A} \subset \mathcal{S}$ is a probabilistic model $\mathcal{M}_{|\mathcal{A}} \equiv (\mathcal{A}, \mathcal{P}_{|\mathcal{A}})$, where $\mathcal{P}_{|\mathcal{A}}$ is the set of probability measures obtained by conditioning every probability measure in $\mathcal{P}$ on $\mathcal{A}$.

Definition 2.7 (model radius). Let $\mathcal{M} \equiv (\mathcal{S}, \mathcal{P})$ be a probabilistic model. We define the *radius* of $\mathcal{M}$ as follows:

$$\mathbb{R}_{\mathcal{M}} = \min_{P \in \mathcal{P}} \max_{Q \in \mathcal{P}} \{\mathbb{D}_{\text{KL}}(Q||P)\} = \min_{P \in \mathcal{P}} \max_{Q \in \mathcal{P}} \sum_{s \in \mathcal{S}} Q(s) \ln \frac{Q(s)}{P(s)} . \quad (10)$$

References

- [1] Nobody Jr. My article, 2006.